

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: SLD
Study Case: LF

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Electrical Transient Analyzer Program

Load Flow Analysis

Loading Category (1): Design
Generation Category (1): Design
Load Diversity Factor: None

	Swing	V-Control	Load	Total
Number of Buses:	1	0	7	8

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	1	0	0	6	0	0	7

Method of Solution: Adaptive Newton-Raphson Method
Maximum No. of Iteration: 99
Precision of Solution: 0.0001000

System Frequency: 50.00 Hz
Unit System: Metric
Project Filename: SLD
Output Filename: C:\Users\ASUS\OneDrive\Desktop\Projects\Udemy\Dialog\interior\SLD\test1.lfr

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Adjustments

<u>Tolerance</u>	<u>Apply Adjustments</u>	<u>Individual /Global</u>	<u>Percent</u>
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		
<u>Temperature Correction</u>	<u>Apply Adjustments</u>	<u>Individual /Global</u>	<u>Degree C</u>
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus1	11.000	1	100.0	0.0								
Bus3	0.400	1	100.0	0.0								
Bus7	0.400	1	100.0	0.0								
Bus9	0.400	1	100.0	0.0								
Bus10	0.400	1	100.0	0.0			0.051	0.032				
Bus12	0.400	1	100.0	0.0								
Bus15	0.400	1	100.0	0.0	0.044	0.027	0.066	0.041				
Bus16	0.400	1	100.0	0.0			0.051	0.032				
Total Number of Buses: 8					0.044	0.027	0.168	0.104	0.000	0.000	0.000	0.000

Generation Bus				Voltage		Generation			Mvar Limits	
ID	kV	Type	Sub-sys	% Mag.	Angle	MW	Mvar	% PF	Max	Min
Bus1	11.000	Swing	1	100.0	0.0					
						0.000	0.000			

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Line/Cable/Busway Input Data

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)									
Line/Cable/Busway									
ID	Library	Size	Length		#/Phase	T (°C)	R	X	Y
			Adj. (m)	% Tol.					
Cable1	1.0MALN3	95	10.0	0.0	1	75	0.376708	0.089000	
Cable3	1.0MALN3	240	30.0	0.0	1	75	0.150880	0.087100	
Cable4	1.0MALN3	16	10.0	0.0	1	75	2.218920	0.103700	
Cable5	1.0MALN3	35	10.0	0.0	1	75	1.015980	0.095500	
Cable6	1.0MALN3	35	12.0	0.0	1	75	1.015980	0.095500	
Cable7	1.0MALN3	240	20.0	0.0	1	75	0.150880	0.087100	

Line / Cable / Busway resistances are listed at the specified temperatures.

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2-Winding Transformer Input Data

Transformer		Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	Phase	MVA	Prim. kV	Sec. kV	% Z1	X1/R1	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
T1	3-Phase	0.300	11.000	0.400	4.00	10.00	0	0	0	0	0	4.0000	Dyn	0.000

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Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
T1	2W XFMR	Bus1	Bus3	132.67	1326.72	1333.33	
Cable1	Cable	Bus3	Bus7	235.44	55.63	241.92	
Cable3	Cable	Bus3	Bus9	282.90	163.31	326.65	
Cable4	Cable	Bus7	Bus10	1386.83	64.81	1388.34	
Cable5	Cable	Bus7	Bus12	634.99	59.69	637.79	
Cable6	Cable	Bus12	Bus16	761.99	71.63	765.34	
Cable7	Cable	Bus9	Bus15	188.60	108.88	217.77	

LOAD FLOW REPORT

Bus		Voltage		Generation		Load		Load Flow					XFMR	
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap	
* Bus1	11.000	100.000	0.0	0.206	0.134	0.000	0.000	Bus3	0.206	0.134	12.9	83.7		
Bus3	0.400	97.978	-1.5	0.000	0.000	0.000	0.000	Bus7	0.097	0.060	168.0	85.3		
								Bus9	0.108	0.067	186.7	85.0		
								Bus1	-0.205	-0.126	354.7	85.1		
Bus7	0.400	97.711	-1.4	0.000	0.000	0.000	0.000	Bus3	-0.097	-0.060	168.0	85.2		
								Bus10	0.048	0.030	84.0	85.2		
								Bus12	0.048	0.030	84.0	85.2		
Bus9	0.400	97.556	-1.5	0.000	0.000	0.000	0.000	Bus3	-0.107	-0.066	186.7	85.0		
								Bus15	0.107	0.066	186.7	85.0		
Bus10	0.400	97.004	-1.2	0.000	0.000	0.048	0.030	Bus7	-0.048	-0.030	84.0	85.0		
Bus12	0.400	97.378	-1.3	0.000	0.000	0.000	0.000	Bus7	-0.048	-0.030	84.0	85.1		
								Bus16	0.048	0.030	84.0	85.1		
Bus15	0.400	97.274	-1.5	0.000	0.000	0.107	0.066	Bus9	-0.107	-0.066	186.7	85.0		
Bus16	0.400	96.979	-1.2	0.000	0.000	0.048	0.030	Bus12	-0.048	-0.030	84.0	85.0		

* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)

Indicates a bus with a load mismatch of more than 0.1 MVA

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Bus Loading Summary Report

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic		MVA	% PF	Amp	Percent Loading
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar				
Bus1	11.000										0.246	83.7	12.9	
Bus3	0.400			0.000							0.241	85.1	354.7	
Bus7	0.400			0.000							0.114	85.2	168.0	
Bus9	0.400			0.000							0.126	85.0	186.7	
Bus10	0.400				0.048	0.030					0.056	85.0	84.0	
Bus12	0.400			0.000							0.057	85.1	84.0	
Bus15	0.400		0.044	0.027	0.063	0.039					0.126	85.0	186.7	
Bus16	0.400			0.000	0.048	0.030					0.056	85.0	84.0	

* Indicates operating load of a bus exceeds the bus critical limit (100.0% of the Continuous Ampere rating).
Indicates operating load of a bus exceeds the bus marginal limit (95.0% of the Continuous Ampere rating).

Branch Loading Summary Report

CKT / Branch		Busway / Cable & Reactor			Transformer				
ID	Type	Ampacity (Amp)	Loading Amp	%	Capability (MVA)	Loading (input)		Loading (output)	
						MVA	%	MVA	%
* Cable1	Cable	157.71	167.99	106.52					
Cable3	Cable	285.94	186.67	65.28					
* Cable4	Cable	50.02	84.01	167.95					
* Cable5	Cable	82.81	83.98	101.41					
* Cable6	Cable	82.81	83.99	101.42					
Cable7	Cable	285.94	186.67	65.28					
* T1	Transformer				0.189	0.246	130.4	0.241	127.7

* Indicates a branch with operating load exceeding the branch capability.

Branch Losses Summary Report

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop in Vmag
	MW	Mvar	MW	Mvar	kW	kvar	From	To	
Cable1	0.097	0.060	-0.097	-0.060	0.3	0.1	98.0	97.7	0.27
Cable3	0.108	0.067	-0.107	-0.066	0.5	0.3	98.0	97.6	0.42
Cable4	0.048	0.030	-0.048	-0.030	0.5	0.0	97.7	97.0	0.71
Cable5	0.048	0.030	-0.048	-0.030	0.2	0.0	97.7	97.4	0.33
Cable6	0.048	0.030	-0.048	-0.030	0.3	0.0	97.4	97.0	0.40
Cable7	0.107	0.066	-0.107	-0.066	0.3	0.2	97.6	97.3	0.28
T1	0.206	0.134	-0.205	-0.126	0.8	8.0	100.0	98.0	2.02
					2.9	8.6			

* This Transmission Line includes Series Capacitor.

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Alert Summary Report

% Alert Settings

	<u>Critical</u>	<u>Marginal</u>
<u>Loading</u>		
Bus	100.0	95.0
Cable / Busway	100.0	95.0
Reactor	100.0	95.0
Line	100.0	95.0
Transformer	100.0	95.0
Panel	100.0	95.0
Protective Device	100.0	95.0
Generator	100.0	95.0
Inverter/Charger	100.0	95.0
<u>Bus Voltage</u>		
OverVoltage	105.0	102.0
UnderVoltage	95.0	98.0
<u>Generator Excitation</u>		
OverExcited (Q Max.)	100.0	95.0
UnderExcited (Q Min.)	100.0	

Critical Report

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Cable1	Cable	Overload	157.705	Amp	167.989	106.5	3-Phase
Cable4	Cable	Overload	50.021	Amp	84.009	167.9	3-Phase
Cable5	Cable	Overload	82.813	Amp	83.983	101.4	3-Phase
Cable6	Cable	Overload	82.813	Amp	83.987	101.4	3-Phase
T1	Transformer	Overload	0.189	MVA	0.246	130.4	3-Phase

Marginal Report

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Bus10	Bus	Under Voltage	0.400	kV	0.388	97.0	3-Phase
Bus12	Bus	Under Voltage	0.400	kV	0.390	97.4	3-Phase
Bus15	Bus	Under Voltage	0.400	kV	0.389	97.3	3-Phase
Bus16	Bus	Under Voltage	0.400	kV	0.388	97.0	3-Phase
Bus3	Bus	Under Voltage	0.400	kV	0.392	98.0	3-Phase
Bus7	Bus	Under Voltage	0.400	kV	0.391	97.7	3-Phase
Bus9	Bus	Under Voltage	0.400	kV	0.390	97.6	3-Phase

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SUMMARY OF TOTAL GENERATION , LOADING & DEMAND

	MW	Mvar	MVA	% PF
Source (Swing Buses):	0.206	0.134	0.246	83.73 Lagging
Source (Non-Swing Buses):	0.000	0.000	0.000	
Total Demand:	0.206	0.134	0.246	83.73 Lagging
Total Motor Load:	0.044	0.027	0.052	85.00 Lagging
Total Static Load:	0.159	0.098	0.187	85.00 Lagging
Total Constant I Load:	0.000	0.000	0.000	
Total Generic Load:	0.000	0.000	0.000	
Apparent Losses:	0.003	0.009		
System Mismatch:	0.000	0.000		

Number of Iterations: 2